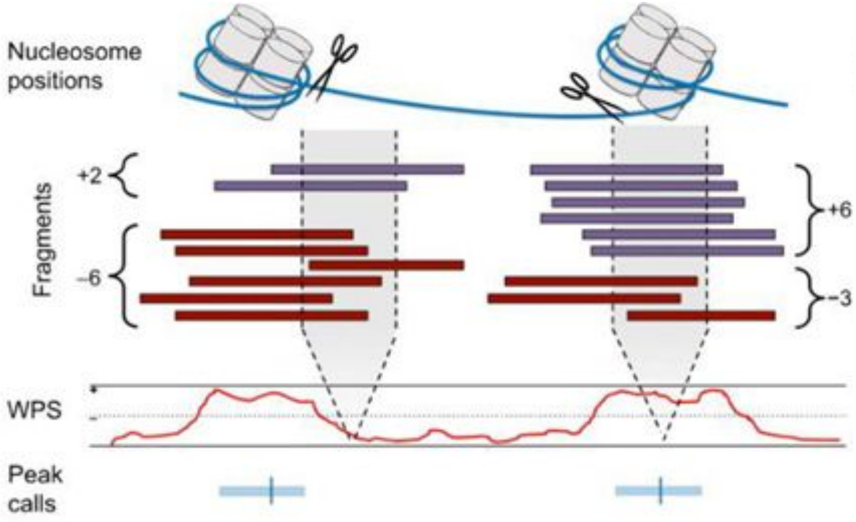
"Classical" Windowed Protection Score (WPS)

$$\mathrm{WPS}_i = (\text{no. spanning fragments}) - (\text{no. ending fragments})$$

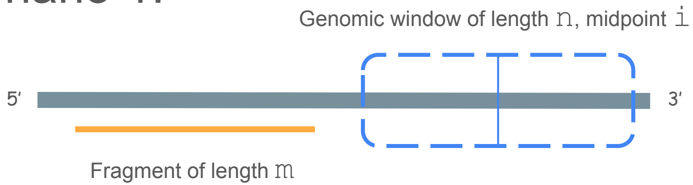


$i = 120$ bp windows along genome
scores are assigned to middle position in window

Snyder, Matthew W., Martin Kircher, Andrew J. Hill, Riza M. Daza, and Jay Shendure. "Cell-Free DNA Comprises an in Vivo Nucleosome Footprint That Informs Its Tissues-of-Origin." *Cell* 164, no. 0 (January 14, 2016): 57–68. https://doi.org/10.1016/j.cell.2015.11.050.

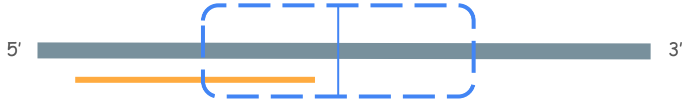
`pyfraglib`'s current implementation

Scenario 1:



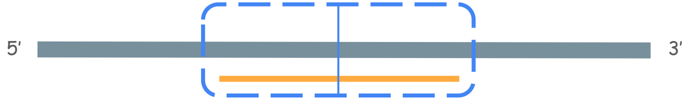
Fragment does not affect WPS at `i`

Scenario 2:



Fragment does WPS at `i: -1`

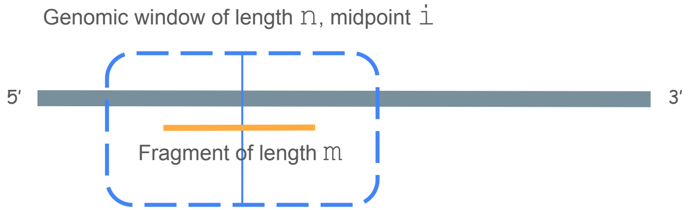
Scenario 3:

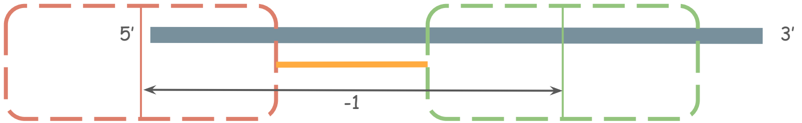


Fragment does WPS at `i: +1`

- Algorithmic idea:
  - for a given fragment, determine all windows that are affected by that fragment
  - Keep a running sum of WPS per site

Case $m<n$:



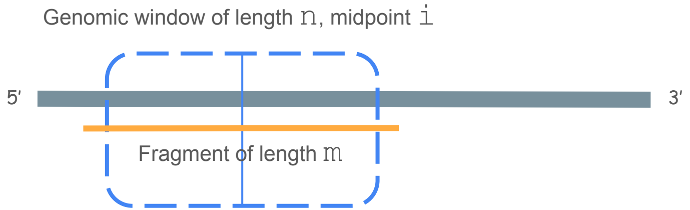


- The midpoint i gets -1 added to its running sum

- In general, all positions between the red and green midpoints get a -1 added because they are midpoints of windows intersecting with the fragment
- This corresponds to the interval `[fragment_start - n/2, fragment_end  + n/2]`

- Algorithmic idea:
  - for a given fragment, determine all windows that are affected by that fragment
  - Keep a running sum of WPS per site

Case `m>=n`:



- `m - n` sites get `+1` added to their running sum, everything else is similar to the first case